

EXAMINATION OF THE EFFECT OF ELEVATED NaCl CONCENTRATIONS ON THE BURDEN OF THE ECTOPARASITES TRICHODINIDS AND DACTYLOGYRIDS ON PIKEPERCH (*SANDER LUCIOPERCA*) MAINTAINED IN RECIRCULATING AQUACULTURE SYSTEMS (RAS)

Harish Muhammad^{ab}, Christopher Naas^{ab}, Andreas Müller-Belecke^a, Werner Kloas^b

^aInstitute of Inland Fisheries Potsdam-Sacrow, Im Königswald 2, D-14469 Potsdam, Germany

^bLeibniz-Institute of Freshwater Ecology and Inland Fisheries, Müggelseedamm 310, 12587 Berlin, Germany



IGB
Leibniz-Institut für
Gewässerökologie
und Binnenfischerei

INTRODUCTION



Pikeperch (*Sander lucioperca*)

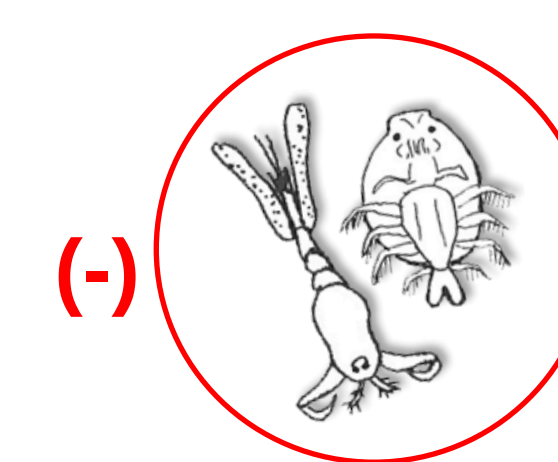
- ✓ High-valued fish in Europe
- ✓ Declining wild catches, but increasing demands

Intensification of
production
technique



Recirculating Aquaculture System (RAS)

- (+) High density culture with less water consumption, smaller space requirement, and in overall lower environmental impact [1]



Risk for parasites
transmission due to
high density [1] [2]



NaCl Salt

- ✓ Easy to apply
- ✓ Provide a positive physiological effect
- ✓ Treat ectoparasites infestation [2] [3]

Aim

To examine the effect of elevated NaCl salt concentrations (0, 3, 6 PSU) to control ectoparasites Trichodinids and Dactylogyrids on pikeperch maintained in RAS

MATERIALS & METHODS

Overwintering Conditions



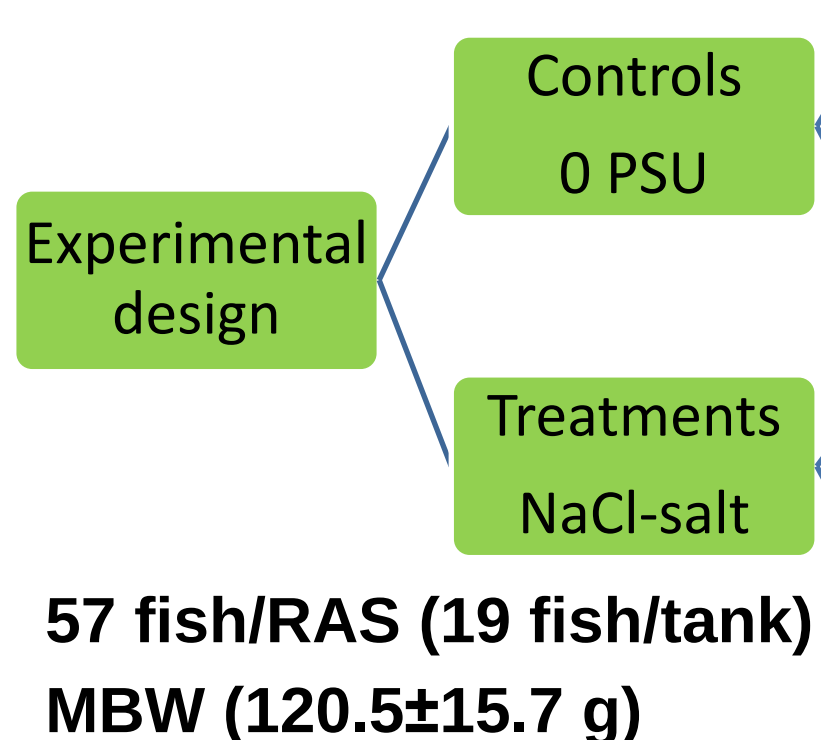
Pikeperch were kept
in lake Sacrow
(August 2018-May 2019)



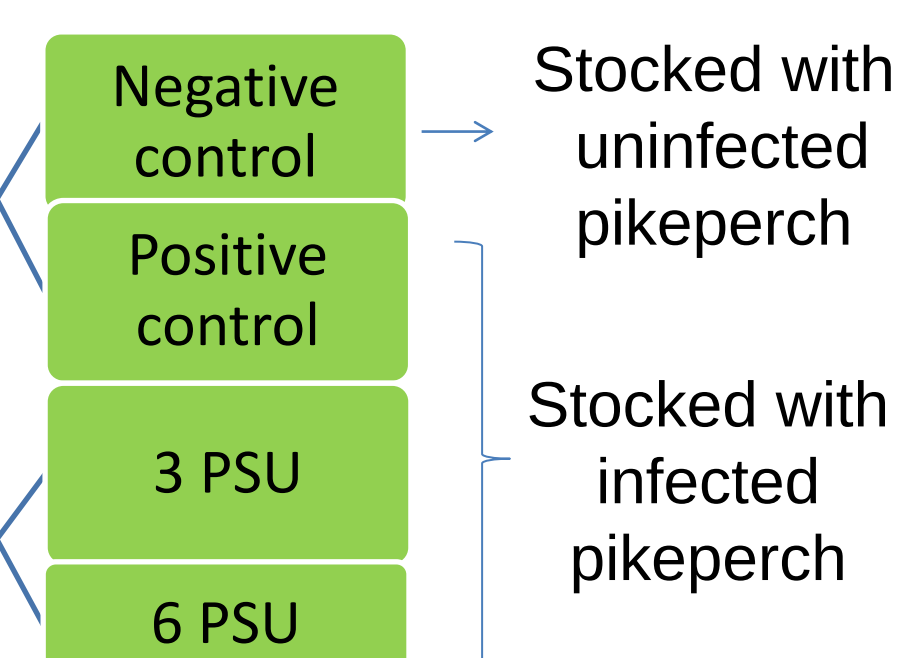
Each RAS contains three
rearing tanks (320 L) and
a filtration unit (120 L)

Experimental Design

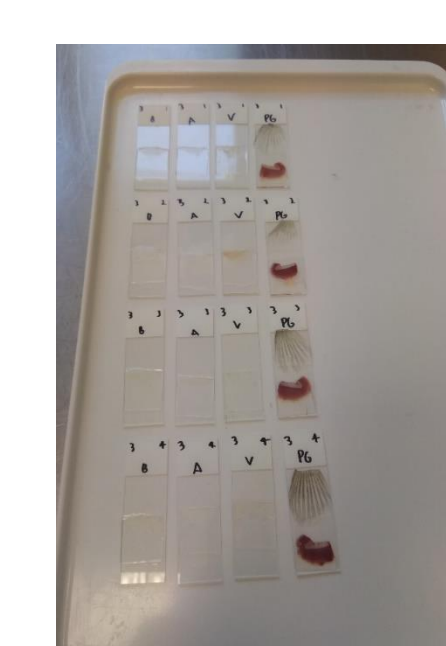
Four units of RAS



21 days experiment



Ectoparasite Sampling



Ectoparasite
Population
Analysis

- Dorsal Fin
- Anal Fin
- Ventral Fin
- Pectoral Fin
- Gills

Sampling time: Day 0, 7, and 21 → 12 fish/treatment

RESULTS

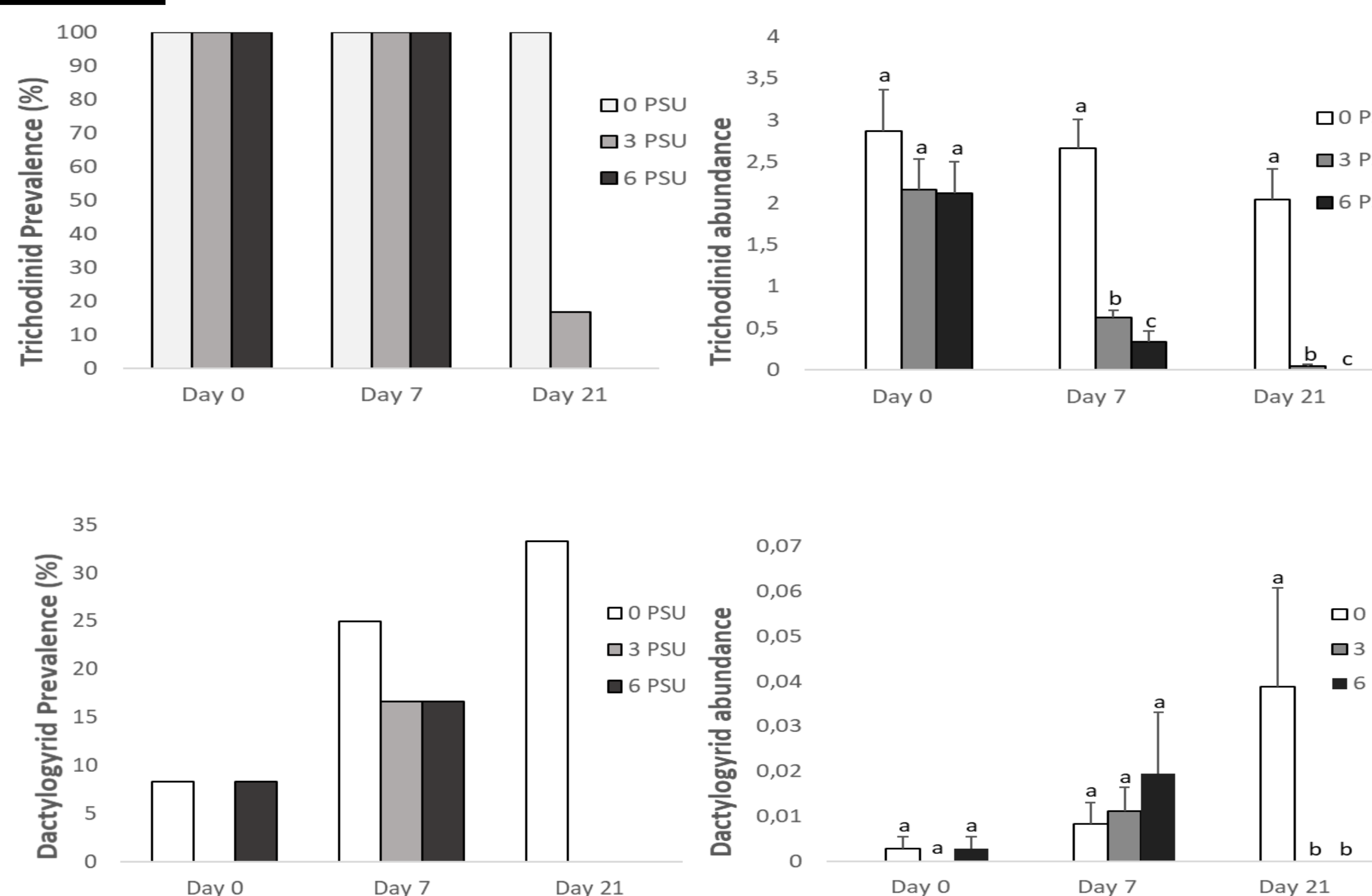
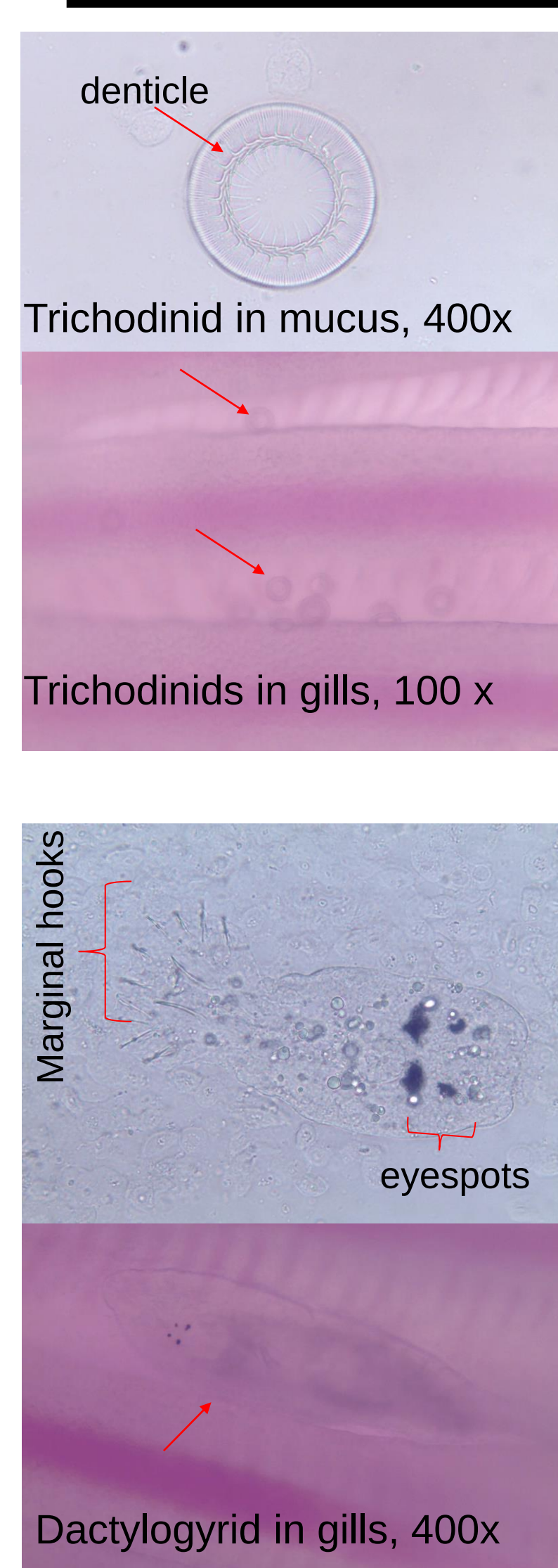


Figure 1. Prevalence and overall abundance of Trichodinids and Dactylogyrids on pikeperch under NaCl saline conditions after 24 h, 7 d, and 21 d (n=12). Values are mean ± SE. Significant differences are marked by lower case letters ($p<0.05$).

DISCUSSION

Treatment with 6 PSU NaCl was the most effective in controlling ectoparasites population by reducing both, prevalence and abundance of Trichodinids and Dactylogyrids.

Concentration of 3 PSU is capable of reducing Dactylogyrids completely and Trichodinids by 83.3%.

Oppositely, there was a stable trend of Trichodinids prevalence (100%) and an increasing trend of Dactylogyrids prevalence (up to 25%) in the positive control (0 PSU).

NaCl salt can remove ectoparasites: By initiating an osmotic pressure difference [4] or by removing excess mucus and debris that are associated with ectoparasite infestations [3]

Conclusion

Despite further research is needed, our findings, especially concerning 6 PSU NaCl, provide a potential effective treatment method for controlling both ectoparasites, Trichodinids and Dactylogyrids, on pikeperch maintained in RAS.

References:

- [1] Bregnballe, J. (2015). A Guide to Recirculation Aquaculture: An introduction to the new environmentally friendly and highly productive closed fish farming systems. FAO and Eurofish Report, 100.
- [2] Németh, S., Horváth, Z., Felföldi, Z., & Beliczky, G. (2013). The use of permitted ectoparasite disinfection methods on young pike-perch (*Sander lucioperca*) after transition from over-wintering lake to RAS. AACL Bioflux 6(1), 1–11.
- [3] Noga, E. J. (2010). Fish Disease: Diagnosis and Treatment, Second Edition (2nd ed.). North Carolina, US: John Wiley & Sons, Inc.
- [4] Kirschner, L. B. (2004). The mechanism of sodium chloride uptake in hyperregulating aquatic animals. Journal of Experimental Biology, 207(9), 1439–1452.

Acknowledgment

This research was financially supported by the German Federal Environmental Foundation (DBU). Harish Muhammad was funded by a scholarship of the Indonesian Endowment fund for Education (LPDP).